

Samit Madatanapalli

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EXPERIENCE

Honeywell

05/2025 – Present

Embedded Engineer 1 (Former Intern)

Pittsburgh, PA

- Developed and deployed a real-time free-fall detection feature in C for Honeywell embedded Linux devices; conducted industrial drop-testing to validate sensor processing accuracy
- Reverse engineered legacy battery charger systems by analyzing MCU-to-battery communication (I2C, UART, 1-Wire) using Saleae logic analyzers and Perl scripts ensuring hardware compatibility for next generation charging systems
- Developed Python and shell scripts to capture Adaptive Frequency Hopping (AFH) data from headset connections and convert logs into 2D channel map visualizations, resolving RF connectivity issues for a major customer
- Engineered a Python telemetry tool to automate metric extraction and visualization from Embedded Linux logs, eliminating manual analysis and accelerating debugging for the electrical engineering team

Flourish: Grow with Self-Care

01/2025 – 05/2025

Intern | Full Stack Mobile Application Engineer

Remote

- Engineered a React Native "Plants Encyclopedia" with custom UI animations and a Python/Appwrite backend to deliver real-time search and personalized content based on user preferences

Persistent Systems / Thermo Fisher Scientific

05/2024 – 08/2024

Intern | Data Engineer Mentee

Remote

- Refactored a personal AI Fitness Tracker using ER/Studio and SQL, implementing normalized database schemas to optimize storage for health metrics and AI-generated nutritional data.
- Applied enterprise-grade data governance standards to ensure data integrity and query efficiency; collaborated with mentors in daily syncs to refine data transformation workflows.

PROJECTS

Esotaira: Military Omnidirectional UAV (Startup)

08/2025 – 12/2025

Capstone | Embedded Software Lead

University Park, PA

- Developed C++ firmware to manage 0–180° tilt-rotor modulation, enabling omnidirectional thrust and high-agility maneuvering for military-spec flight profiles
- Engineered the full electrical architecture, implementing high-reliability wire routing and signal-integrity hardening (soldering, heat-shrinking) for ruggedized use
- Optimized Onyx 3D-printed airframe components and assisted in the construction of an epoxy-glass testing enclosure to ensure structural integrity during high-torque stress tests

Offline AI-enabled Drone

09/2025 – 12/2025

Embedded Lead

University Park, PA

- Engineered a custom quadrotor drone that broadcasts live aerial telemetry to a localized Raspberry Pi 5 ground station for high-speed field surveillance.
- Developed an offline detection pipeline using YOLOv8 and OpenCV to identify personnel and vehicles in real-time, overlaying dynamic AI labels onto a mobile touchscreen.

SKILLS

Software: C/C++, Python/Shell automation, Embedded Linux (Yocto), Service files, Bootloaders, RTOS, ISRs, Memory Management, Verilog, Vivado, SQL, HCS12 Assembly, Git, Serial Communication (I2C, SPI, UART), Socket Programming

Hardware: Cadence, MATLAB; RISC-V, MCUs; Oscilloscope Waveform Analysis, Saleae Logic Analyzer, Function Generators; Circuit Design & Analysis, Power Management, Sensor Integration (ADC, PWM, Timers, GPIO), Datasheet Reading, Soldering

EDUCATION

The Pennsylvania State University | College of Engineering

Graduated: 05/2026

- B.S. in Computer Engineering
- Dean's List 2023-2024-2025

University Park, PA
GPA: 3.52